

ACTION CHUNK SEAM TIMELINE & FALLBACK DECELERATION DYNAMICS

Temporal Milestones, Replacement Ingestion Windows, and Clearance-Bounded Braking Contracts



REGIME 1 ON-TIME ARRIVAL $t_{arr} \leq t_{blend}$

Active Chunk A Execution ($t \in [0, 300 \text{ ms}]$)

100 ms Blend

Chunk B Continuous Execution →

✓ Inference arrives at P50 = 180 ms. Full 100 ms blend window available. Smooth C² quintic interpolation preserves continuous motion.

REGIME 2 LATE-BUT-BLENDBLE $t_{blend} < t_{arr} \leq t_{commit}$

Chunk A Latency Extension ($t_{arr} = 340 \text{ ms}$)

60 ms Blend

Chunk B Continuous Execution →

⚠ Tail latency delay (P99 = 340 ms). Blend window compressed from 100 ms to 60 ms. Peak acceleration check required: $a_{peak} \sim 6\Delta q/T_{blend}^2$. If $a_{peak} \leq a_{max}$ (6.0 m/s²), admit compressed blend; if $a_{peak} > a_{max}$, abort to Fallback Suffix immediately.

REGIME 3 FALLBACK SUFFIX $t > t_{commit}$ / Lost

Active Chunk Execution to Commitment ($t_{commit} = 360 \text{ ms}$)

PRECOMPUTED FALLBACK DECELERATION (200 ms Brak) REST

Distance Traveled: $d_{total} = d_{late} (72 \text{ mm}) + d_{brake} (120 \text{ mm}) = 192 \text{ mm} < d_{clear} (300 \text{ mm})$

🔴 At $t_{commit} = 360 \text{ ms}$, replanning closes. Controller branches into deterministic C² braking at $a_{max} = 6.0 \text{ m/s}^2$, bringing mechanism to zero velocity at $t_{term} = 560 \text{ ms}$ with 108 mm remaining margin. Mechanical holding brakes engage; zero current draw.

PHYSICAL TAXONOMY: WHY DEFAULT "HOLD" BEHAVIORS FAIL AT SEAM EXPIRY

1. Hold Commanded Torque

- Maintains constant motor current
- External contact / gravity changes
- Mechanism accelerates uncontrolled

Result: Runaway joint acceleration
Thermal runaway in coils

2. Hold Position Setpoint

- Commands $v = 0$ instantaneously
- Step velocity jump at moving tool
- Inverter overcurrent trip

Result: 324 N·m torque shock
Gear tooth brinelling & trip

3. Continue Commanded Velocity

- Avoids transient current spike
- Expendes clearance at 1.2 m/s
- Traverses 300 mm gap in 250 ms

Result: Fixture hard impact
Tool breakage & payload drop

4. Verified Fallback Suffix

- Precomputed C² deceleration
- Budgeted stopping dist: 192 mm
- Resident in memory at $t = 0 \text{ ms}$

Result: Safe stop at t_{term}
Mechanical brakes engage safely